

Revenue-at-Risk Retention Analysis

A Case Study in Churn Prediction & Business Decision-Making

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The question: which currently active customers are most likely to churn next, and how much revenue does that represent - so a retention team knows exactly who to call first, and can defend the dollar case for doing it.

7,032

historical customers
analyzed

0.833

held-out test AUC

\$114.6K

expected revenue at risk,
top 150 accounts

49%

of revenue lost to churn
from one cohort

1. Executive Summary

Retention outreach at Meridian Communications (*a fictional company built on real, public data - see the Data section*) was reactive rather than prioritized: there was no systematic way to identify which active customers were most likely to churn, or how much revenue was actually on the line. This project builds that system end to end: a validated statistical model of churn drivers, a forward-looking risk score applied only to currently active accounts, and a dollar-ranked call list a retention team could act on immediately.

The single biggest finding: **customers on month-to-month contracts in their first year account for 49% of all annualized revenue lost to churn**, despite being a minority of the customer base. The top 150 currently active accounts ranked by the model carry \$114,608 in expected annualized revenue at risk. On customers the model never saw, the accounts it ranked in its top 10% actually churned at 72.3% versus 26.6% overall - a 2.7x lift - which is what makes the ranked list worth acting on.

2. Business Context

A subscription telecom/internet provider loses revenue every time a customer churns, but not every churned customer is equally costly, and not every active customer is equally at risk. Without a model, retention teams either contact everyone (expensive, low signal) or rely on gut feel about which segments are risky (imprecise, not dollar-weighted). The goal here is a specific, defensible answer to two questions a VP of Customer Success would actually ask: *who should we call this month, and what's the dollar case for calling them instead of someone else?*

3. Data

The analysis uses IBM's public Telco Customer Churn dataset (7,043 customers, 21 attributes covering tenure, contract type, services subscribed, billing method, and churn outcome). It is framed here as a fictional company, "Meridian Communications," to structure the work as a real retention decision rather than a data-exploration exercise - every number in this document is computed from the real dataset, not fabricated.

4. Methodology

4.1 Data Cleaning

- Excluded 11 customers with zero tenure - they have no billing history yet and cannot meaningfully be scored for churn risk.
- Collapsed the "No internet service" category (present in six add-on columns: online security, backup, device protection, tech support, streaming TV/movies) into "No." Left uncollapsed, this category is perfectly collinear with *InternetService = No* and produces a singular matrix in logistic regression - a real bug encountered and fixed during this analysis, not a hypothetical.
- Bucketed tenure into four cohorts (0–12, 13–24, 25–48, 49–72 months) for cohort-level reporting.

4.2 Statistical Testing

Chi-square tests of independence were run for eight categorical variables against churn (Contract, InternetService, TechSupport, OnlineSecurity, PaperlessBilling, PaymentMethod, SeniorCitizen, Dependents). All eight are significant at $p < 0.001$, confirming they are not independent of churn before any modeling assumptions are introduced.

4.3 Predictive Modeling

A logistic regression was fit on the full historical dataset (tenure, monthly charges, contract, internet service, tech support, online security, paperless billing, payment method, senior citizen status, dependents) to identify which factors drive churn and in which direction, with p-values for statistical significance. Separately, a scikit-learn logistic regression was trained on an 80/20 train/test split for the prediction task itself, and evaluated on the held-out 20% - achieving an **AUC of 0.833**, meaning the model meaningfully separates churners from non-churners rather than performing close to chance (0.5).

4.4 Scoring & Ranking

The model was refit on the complete historical dataset, then applied only to the 5,163 **currently active** customers - a customer who has already churned does not need a future-risk prediction. Each active customer was scored with: *expected revenue at risk* = *churn probability* × *annualized revenue* (monthly charge × 12). This is a probability-weighted expected value, not a customer's actual billing - a distinction worth stating explicitly, since it changes how the top-line dollar figure should be interpreted (see Section 5.3).

5. Findings

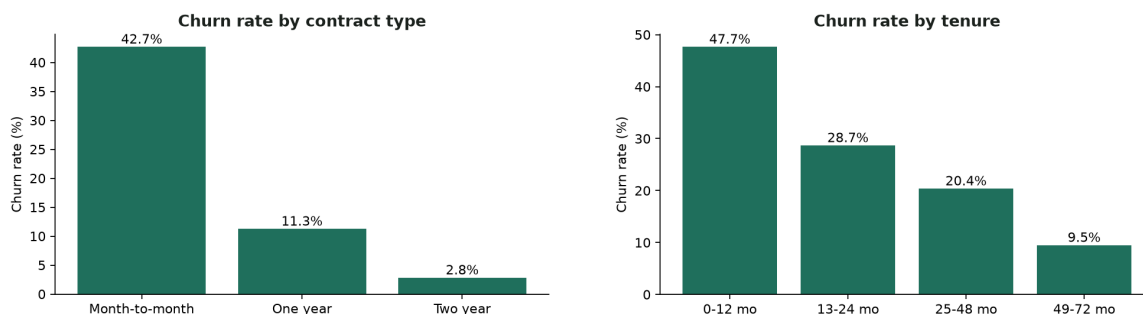
5.1 What Actually Drives Churn

Ten variables are statistically significant churn predictors at $p < 0.001$. They split cleanly into protective and risk-increasing factors:

Factor	Effect	Odds Ratio	p-value
Two-year contract (vs. month-to-month)	Protective	0.267	< 0.001
One-year contract (vs. month-to-month)	Protective	0.518	< 0.001
Longer tenure (per month)	Protective	0.969	< 0.001
Has online security add-on	Protective	0.635	< 0.001
Has tech support add-on	Protective	0.679	< 0.001
No internet service	Protective	0.465	< 0.001
Fiber-optic internet	Increases risk	1.658	< 0.001
Pays by electronic check	Increases risk	1.431	< 0.001
Paperless billing	Increases risk	1.449	< 0.001
Higher monthly charges (per \$)	Increases risk	1.011	< 0.001

Contract length has the largest effect of any categorical driver: moving from a month-to-month to a two-year contract reduces the odds of churn by roughly 73%, holding other factors constant. Two secondary signals were also significant at a weaker threshold ($p < 0.05$): having dependents is mildly protective (OR 0.838), and being a senior citizen mildly increases risk (OR 1.286).

5.2 Where the Charts Confirm the Story



Churn falls from 42.7% (month-to-month) to 2.8% (two-year contracts), and from 47.7% in a customer's first year to 9.5% after four years - both consistent with the regression's conclusions above, not just the raw correlations behind them.

5.3 The Concentration Finding

New, month-to-month customers (0–12 months tenure) are 1,994 of the 7,032 total customers (28%) - but they account for **49.1% of all annualized revenue lost to churn** (\$819,617 of the total). This single cohort is the highest-leverage place to intervene.

5.4 The Ranked Call List

Scoring the 5,163 active customers and ranking by expected revenue at risk surfaces the top 150 accounts to prioritize. A precise reading of the dollar figure matters here: these 150 accounts carry **\$114,608 in expected annualized revenue at risk** (their churn probability weighted against their revenue). Their actual combined billing is higher - **\$167,537/year** - but not all of that is expected to be lost; \$114,608 is the risk-adjusted figure a retention budget should be sized against. Notably, **100% of the top 150 are on month-to-month contracts**, which is not a coincidence given the driver analysis above - it is the model converging on the same conclusion from a different angle.

6. Recommendation & What It Changes

Replace untargeted outreach with the ranked call list: route the top 150 accounts to retention this month, and lead with a free 3-month tech-support or online-security add-on - both are among the strongest statistically significant protective factors - rather than a blanket discount, which the driver analysis does not support.

The evidence for the change, separated by how much each piece can be trusted:

Evidence	Result	Basis
Churn in the model's top-scored 10% vs overall	72.3% vs 26.6% (2.7x)	Measured, held-out customers
Share of all real churners in that top 10%	27.3% (random: 10%)	Measured, held-out customers
Expected at-risk revenue, top 150 vs a random 150	\$114.6K vs \$23.8K (4.8x)	Model-estimated
Revenue retained if the offer saves 10%	\$11,461/yr; offer must cost < \$76/account	Assumption, not a result
Revenue retained if the offer saves 20%	\$22,922/yr; offer must cost < \$153/account	Assumption, not a result
Revenue retained if the offer saves 30%	\$34,382/yr; offer must cost < \$229/account	Assumption, not a result

No retention campaign was run, so the shaded rows are arithmetic on stated assumptions, not outcomes. Replacing them with a measured save rate is the point of the A/B test proposed below.

7. Assumptions & Limitations

- The model was trained on historical outcomes (including customers who already churned) and then applied to active accounts going forward - it has not yet been validated against the actual outcome of a real retention campaign. The recommended offer should be tested on a subset before a full rollout.
- Active customers also appear in the training data labeled as “retained,” although their future is unknown - a standard simplification with snapshot data. A time-based split would be a stronger design.
- Revenue at risk is annualized at the customer's *current* monthly rate; it does not account for future price changes, upgrades, or downgrades.
- Collapsing “No internet service” into “No” for add-on columns removes exact multicollinearity, but a partial correlation remains between InternetService and the add-on columns - coefficients for those add-ons should be read as “effect among customers with comparable internet service,” not fully independent effects.

8. Technical Appendix

Stack: SQL (SQLite) · Python (pandas, statsmodels, scikit-learn) · Power BI

Two SQL approaches, deliberately: alongside the Python model, a pure-SQL rule-based risk heuristic (window functions, no ML dependency) solves the same ranking problem for a team without a data-science function - useful context for explaining when each approach is the right tool.

Full code, data, and the interactive Power BI dashboard:

github.com/parulpatil02-lgtm/retention-risk-analysis